

Ashwath Karthikeyan

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EDUCATION

University of Illionis at Urbana - Champaign
Master's of Engineering in Robotics

Champaign, IL
Graduation Date: Dec 2024

EXPERIENCE

United Parcel Service

Warehouse Robot Data Analyst Co-Op

Alpharetta, GA
Jun 2024 - Present

- Created a data visualization tool to monitor about 200 active warehouse robots simultaneously, yielding insights on various inbound and outbound tasks while reducing incident response time by identifying bottlenecks in processes.
- Improved visibility into robot operations in human-inaccessible areas by analyzing real-time data from the Warehouse Management System, identifying inefficiencies, and enabling a 30% reduction in response times to potential issues and delays.
- Devised an anomaly detection system to identify orders falling below a specified performance threshold, analyze patterns contributing to delays, and predict the source of issues with 80% accuracy, enhancing overall operational insights.

Stellantis (Previously Fiat Chrysler Automobiles)

Systems Engineering Intern

Chennai, India
Jun 2022 - Aug 2022

- Constructed comprehensive system models using IBM Rational Rhapsody, transforming 40+ complex vehicle Functional Requirement Diagrams into structured designs, reducing simulation preparation time by 20 hours per project.
- Built and executed 50+ robust test scenarios to validate system model accuracy, ensuring alignment with 10 critical functional and operational requirements while identifying improvements to enhance reliability and performance.
- Secured executive approval to deploy thoroughly validated models, enabling seamless integration into 5 downstream processes and saving approximately 200 hours of fabrication rework annually through enhanced efficiency and precision.

PROJECTS

Multirobot Warehouse Automation

ashwath.net/multirobot/

Alpharetta, GA
Dec 2024 - Jan 2025

- Simulated multi-robot path planning scenarios in PyGame, achieving optimized simultaneous robot movements across complex warehouse grids, reducing average collision rates by 35% through an innovative deadlock-prevention exchange maneuver.
- Automated algorithmic performance evaluations by comparing deterministic methods like Dijkstra and A* with probabilistic approaches, benchmarking solution efficiency and execution time across diverse warehouse layouts.
- Orchestrated simultaneous task completion for up to 10 robots in simulated environments, achieving 99% obstacle-avoidance accuracy and reducing task resolution times by 20% compared to baseline methods, ensuring efficient multi-robot coordination.

Package Detection and Tracking on Autonomous Drone

ashwath.net/drone/

Alpharetta, GA
Jun 2024 - Sep 2024

- Implemented an autonomous drone system to detect cardboard boxes in the camera's field of view and navigate towards them, operating reliably within a 20-foot range and achieving a swift response time of under 200 milliseconds.
- Trained a deep learning model for real-time detection of cardboard boxes, accurately determining their position relative to the drone using bounding box coordinates, achieving an accuracy of 93% in various operational environments.
- Programmed an algorithm to maneuver the drone towards the detected box, ensuring precise proximity navigation. Once within range, implemented barcode scanning with a 98% decoding accuracy, enabling efficient data retrieval for automated workflows.

Automatic Parking in Self-Driving Car

ashwath.net/self-parking/

Champaign, IL
Jan 2024 - May 2024

- Developed a full autonomy stack for a self-parking functionality on an autonomous vehicle, enabling real-time detection of empty parking spots and precise navigation of the vehicle into the detected parking spot with a success rate of 70%.
- Designed and tuned an MPC controller for steering and velocity control, ensuring precise trajectory tracking at speeds up to 20 mph. Programmed a Hybrid A* algorithm for efficient real-time route planning with optimized weights.
- Modeled a robust vision-based deep learning system capable of identifying cardboard boxes in real time, reaching processing speeds of 30 FPS while maintaining precise bounding box coordinates to optimize drone performance.

SKILLS

Proficiencies: Robot Operating System (ROS), Python, OpenCV, SolidWorks

Domains: Warehouse Management Systems (WMS), Warehouse Robotics, Autonomous Vehicles,